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Integrating Mparse constraint into output-to-output correspondence

-a critical review on *Less than zero: Correspondence and the null output*
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1. Introduction

Paradigmatic gap, a phonological absence of morphological variation, has long threatened the fundaments of the OT framework. If the output from the EVAL only incurs the least but is not entirely immune to violations, there was no reason to completely rule out every possible candidate. In other words, paradigmatic gap undermined the violable constraint assumptions in OT. One simple coping mechanism was to make the gap itself an output candidate. Coined *null parse* (Prince & Smolensky 1993), this phonologically empty option was stipulated to be in obedience to all faithfulness and markedness constraints to the exception of Mparse. Mparse constraint was another newly published theoretical device that judged the feasibility of parsing. The emptiness of null parse ensured the infeasibility of parsing as opposed to its counterparts with phonetic substance. As a result, with Mparse ranked sufficiently low, empty null parse could get selected since the only violation it incurs concerned the low-ranked Mparse.

	CON (markedness & faithfulness constraints)	Mparse
Candidates with phonetic realization	* (is likely to violate at least one faithfulness or markedness constraints in this set)	
☞ Null parse		*

While the exclusive interplay between these newly coined concepts offered analytical stability, their identities were yet unclear. The key issue left unattended was how we can differentiate null parse from an output of complete deletion. This problem especially loomed large when parsing was defined in phonological domain. Alternatively pertaining Mparse to morphological domain and relating it to parsing of its content could save the dynamic, but it nevertheless begs the question of how null parse can escape phonological penalty. What was needed seemed to be entire rearrangement of correspondence relation where Mparse can be evaluated on.

This is the context where McCarthy & Wolf (2005) brings in the idea of string-based correspondence via concatenative decomposition of the input and output form. If the output resulting in deletion of all segments were not to violate Mparse, the deleted trace should still remain as valid candidate for parsing. The flip side of postulating this is to also allow an input equivalence of parse-feasible null trace. The

input null trace accounts for the epenthesis procedure, so that both deletion and epenthesis can now participate in one-on-one correspondence relation as follows:
(# stands for null trace whereas Sn stands for the nth string)

Input / s1, s2, s3, #/ Input /s1, s2, s3, s4/
Output /s1, s2, s3, s4/ <epenthesis> Output / s1, s2, s3, #/ <deletion>

With this revised definition of epenthesis and deletion, all conventional phonological procedures can now be assessed under bijective relation. Mapping from trace# to phonetically realized string does not amount to correspondence failure, which can only be caused by having no correspondent at all. Unlike #, a complete empty presence as in /s1,s2,s3,s4/ →/s1, s2, s3/ gets penalized, where the relationship input string s4 form with the output is undefined. Meanwhile, s4 does cannot incur faithfulness violation (deletion) with its undefined mapping. The given logic separates two important concepts that are the building block of defining null parse and Mparse: 1) null string # from empty presence and 2) faithfulness assessment from correspondence assessment.

Mparse is a constraint specialized in assessing whether the correspondence is bijective. Null parse is an instance when the output is empty and the relationship is undefined. With these assumptions, null parse meets the goal of violating only Mparse. It first escapes the markedness evaluation as the output is a complete empty presence. Secondly, it eschews the faithfulness judgement with the undefined relationship. With vacuous satisfaction of all faithfulness and markedness constraint, null parse should be chosen as the optimal candidate among those that violate constraints ranked above Mparse. In formal notation it writes as; ([],<i>,<>,R=∅), borrowing the definition of concatenative decomposition.

In line with defining null trace #, both input <i> and output <o> are free to be decomposed into strings of empty to multi segments. The same word apple, for instance, can be decomposed into <a,pp,l,e>,<a,p,p,l,e>, or <app,#,le> even. The <i> notation shows that the decomposition is occupied while the <> notation shows the unoccupied output decomposition. [] stands for empty final output and ∅ stands for undefined relation R. We will revisit this concept in more detail in the following section, but for now it is important to note how the correspondent unit has changed from a segment to a string. This change in machinery affects correspondence relations in the input-output domain and output-to-output domain alike. While it offers expository benefits in the former, its expansion to the latter proves troublesome. In this paper, we will delve into the troublesome nature of Mparse in output-to-output correspondence and ultimately call for a revision to overcome the stated issues. It

will thus be organized as follows. Section 2 will cover how string-based correspondence successfully explains the gap in the input-output domain and why the categorical nature of Mparse is the prerequisite to be so. Section 3 will explore its expansion to the output-to-output domain and point out the problem that arises in both base-reduplicant and between free-standing outputs relations. Section 4 will attempt to identify the source of the issue and make revisions for output-to-output Mparse to be gradient. Lastly, section 5 will compare alternative approaches to Mparse, and show how our revision helps the theory surpass its competitors.

2. Application in Input-Output domain

One example of the input-output gap can be found in Turkish suffixation. In Turkish word formation, bimorphemic words should at least be disyllabic. Accordingly, affixation of the singular genitive suffix *-m* results in a gap for a vowel ending monosyllabic root as in *do* (translates to: musical note C) ; **do: -m* (my C). A relevant well-formedness constraint at play will be *bimorphemic σσ, while competing faithfulness constraint DEP could satisfy the requirement by inserting an epenthetic vowel to break the monosyllabic sequence. The selection of null parse insinuates the dominating two forces towards Mparse as follows: DEP, *bimorphemic σσ >> Mparse

/do: -m/	*bimorphemic σσ	DEP	uniformity	Mparse
a. [do:m] <i> = <d, o:. m> <o> = <d,o:,m> R = {(d,d),(o:,o:),(m,m)}	*			
b. [dou:m] <i>=<d,o:,#,m> <o>=<d,o:,u,m> R={(d,d),(o:,o:),(#,u),(m,m))		*		
c. [do:] <i>=<d,o:m> <o>=<d,o:> R={(d,d),(o:m,o:)}			*	
d. [do:] <i> = <d,o:,m> <o>=<d,o:> R={(d,d),(o:,o),R(m) is undefined}				*
e. [] <i> = <d,o:,m>				*

<o> = <> R=∅				

For expository purposes, I have also added a uniformity-violating option c to show how multi-segmental string correspondence works. The ranking of the three markedness and faithfulness constraints is not too relevant for our discussion; hence the equal ranking position. Comparing e from a and b, we can see how separating correspondence constraint Mparse from faithfulness constraints renders Mparse to be the only violating constraint of null parse e. Option e cannot violate markedness and faithfulness constraints as it lacks both output form and mapping relation to judge them against. In addition, although c and d share the same surface-level outcome, string-based correspondence incurs less violation for a partially defined relationship than for coalescence.

A more important comparison lies between d and e. Both have satisfied three higher-ranked constraints above Mparse, yet the degree of parsing violation differs. Among decomposed elements of the input, d has only one element failing to form correspondence, whereas e has all three failing to do so. If Mparse assigned violations for every non-defined relationship of each string, d would win over e, against the attested findings. For null parse e to be chosen in case of failure to meet the disyllabic condition, we would need to assume a categorical nature of Mparse. Whether the undefined realm spans the entire input strings or is limited to a few, only a single Mparse violation must be assigned for failing to upkeep the bijectivity. This additional stipulation essentially stems from a need for the harmonic bounding status of the null parse. Once Mparse becomes gradient, null parse runs the chance of being suboptimal to candidates with partially defined relationships. While this precondition is indispensable in input-output relation, we shall later see that it is not so necessary for an output-to-output correspondence.

Another important revelation from this ranking schema is that Mparse exerts a thresholding effect. Candidates that violate constraints ranked above MParse will be robbed of chances to appear as there is always a less costly option to forgo any phonetic realization. As harmonically bounding Mparse violator null parse manifests absolute ungrammaticality, constraints that entail larger violation costs than Mparse become inviolable. This finding can be schematized as follows: inviolable constraints >> Mparse >> violable constraints.

3. Application in output-to-output domain

McCarthy & Wolf (2005) deals with two types of output-to-output correspondence, namely those between base and reduplicant and two free-standing forms. Correspondence relation is a domain-agnostic concept that should apply across various dimensions. Although the fundaments of establishing string-

based correspondence between two representations remain the same, evaluation constraint Mparse itself may vary. For BR-correspondence, the authors claimed no particular issue with using Mparse-BR on top of the concatenative decomposition of base <b> and reduplicant <r>, for which I claim otherwise.

To begin with, if base is a null parse, we could spare the calculation as faithfulness mapping is undefined for a relationship between two empty presence (i.e. ([] , <>, <>, R=∅). Markedness constraint is also vacuously satisfied since the mapping from an empty domain can only result in equivalence. When the base is occupied, however, it becomes troublesome to define Mparse – IO >> M / F >> Mparse – BR relation that either leads to 1) empty reduplicant – non-empty base duo or 2) emergence of the least phonologically marked reduplicant that shares minimal resemblance with the base.

(1)

/RED-pam/	MAX-BR	NO-CODA	Mparse-BR
a. [pam-pam] <b> = <p, a, m> <r> = <p, a, m> R ={(p,p),(a,a),(m,m)}		**!	
b. [pa-pam] <b> = <p, a, m> <r> = <p, a, #> R = {(p,p), (a,a), (m,#)}	*!	*	
c. [pa-pam] <b> = <p,a,m> <r> = <p,a> R= ((p,p),(a,a), R(m) is undefined)		*	*
d. [pam] <b> = <p, a, m> <r> = <> R = ∅		*	* (→ *** after revision)
e. [ʔə-pam] <b> = <p, a, m> <r> = < ʔ, ə > R = ∅		*	* (→ *** after revision)

Assuming higher ranked MParse beyond MAX-BR, the base /pam/ is present with phonetic substance. Thus, the above example is left with 3 indifferentiable options with varying choices of reduplicants. Option c has phonetic realization of /pa/, d has none, and e has minimally marked /ʔə/. In theory, d and e outperforms c, since /pa/ is more marked than /ʔə/. The problem is that the desirable outcome is rather the

opposite. The identity of d is unsettling because it juxtaposes two categorically different correspondents that have been set to be construed in different terms. The core idea behind differentiating null parse from maximal deletion rests on the phonological definability between these two. Null parse was an empty presence that could not be phonologically defined and parsed. Null trace, on the other hand, was a parse-feasible phonological feature. The fact that these two features from different realms are conjugated speaks of their compatibility, which in turn is contradictory. To maintain theoretical coherency, these two should not coexist. A similar role of blocking reduplication gap should be rather taken up by the faithfulness MAX constraint that deals with a more compatible duo; null traces # with phonologically present base.

Option e gives out a more straightforwardly bizarre outcome. Having a minimally marked reduplicant structure that bears little to no resemblance with base is a rarely attested phenomenon. Even if one were to claim it as the emergence of the unmarked, the relationship between /ʔə/ and /pam/ is too arbitrary. In the end, partial reduplication in c seems as the most viable option that does not jeopardize the established theoretical framework, while maintaining a comparatively tolerable amount of phonological resemblance. The next section will look into the amendment that is needed to arrive at this result.

Meanwhile, when the problems of Mparse-BR are unattended, McCarthy & Wolf (2005) drastically negate the Mparse employed among two free-standing forms. Their reason for negation stems from the unique status of representations participating in correspondence. The free-standing outputs each have their own IO relation configured before participating in output-to-output correspondence. Therefore, unlike the BR and IO relation generated by GEN, the OO relation is supposedly established through being computed by it. The logic extends to the fixed status of their relationship that forgoes the need to be evaluated at EVAL. Without the need for this evaluation, Mparse is no longer needed. Nevertheless, while acknowledging this fixed status, I would like to claim a need for the correspondence evaluation. Before elaborating on this detail, I would like to clarify some of the potentially confounding terminologies used in this paper. Output-to-output correspondence is an umbrella term subsuming both base-reduplicant and free-standing output form relations. While each relation is tied to Mparse-BR and Mparse-OO, O-O-Mparse will notate shared characteristics between these two constraints.

4. A call for a revision on O-O-Mparse

So far, we have identified problems associated with current definitions of Mparse on output-to-output correspondence. In BR-relation, it could lead to an unlikely outcome that undermines our theoretical assumption, and results in an unattested phenomenon. Mparse-OO is jeopardized by an existential crisis.

The problem with BR stemmed from penalizing the bijective violation equally regardless of how many strings lost correspondence. If (1)d and (1)e were more heavily penalized than (1)c, we could subvert the results and select the theory-affable option (1) c. As for Mparse-OO, the negation overlooks the importance of attending to the details of the computed relation. The fact that the relationship between two outputs is given does not reveal what modes of connection strings are under. Disregarding this aspect is analytically imbalanced, given the emphasis on stating relation R explicitly in other modes of correspondence. Moreover, since Mparse judges bijective relations in particular, there is a need to distinguish bijective from non-bijective. In a base-truncation relationship, for instance, the non-truncated portion of the base can either correspond to null string # or have its relationship undefined.

/lari/	MAX	MParse-OO
<o1> = <I,a,r,i> <o2> = <l,a,r> R = {(l,l),(a,a),(r,r), R(i) is undefined}		*
<o1> = <l,a,r,i> <o2> = <l,a,r,#> R = {(1,1), (a,a), (r,r), (i,#)}	*	

The common ground between these two shortcomings is that a lack of attention is being paid to the amount of correspondence violation contained within the candidate. Ironically, output-to-output correspondence is where finer-grained details of parsing are the most deserving. OO correspondence differs from IO correspondence in that it is particularly driven by the desire to match identities. Unlike IO correspondence where the input enters to form the final output, output-to-output correspondence forms a relation to maximize the preservation of identity. If the former is a bottom-up procedure to build the final craft with ingredients provided at the input, the latter is a combined bottom-up & top-down process. There are block-building characteristics such as when copying the base to form the reduplicant or referencing the base to create the affixed form. At the same time, however, the act of maximizing the resemblance is that of a top-down adjustment. Such bilateral intricacy makes evaluating every correspondence violation individually important. We thereby earn, an interpretive justification to make the O-O-Mparse gradient.

Perhaps more importantly, though, there is executive justification for this change. Earlier in section 2, the need for categoricity of Mparse in the input-output domain was stated to be the harmonic bounding of null parse to other non-bijective relationships. The reason why Mparse-IO had to be categorical was to make null parse the most optimal Mparse-IO violator. In output-to-output relation, such harmonic bounding requirement is no longer necessary. In free-standing OO correspondence that judges the already present

two IO relations, a null parse is no longer a possibility. In reduplication which is the limitless resource for individual creativity (Tannen 1987), the imperativeness of the gap is less strict. Hence, we may loosen up the categorical requirement and allow variations of violation marking. The definition of Mparse-IO and O-O-Mparse will then go as follows:

MPARSE – IO	O-O-MPARSE
Given a candidate ([o], <i>, <o>, R), if R is not a total bijective function from <i> to<o>, assign a violation-mark.	Given a candidate ([o1(o2)], <o1>, <o2>, R), if R is not a total bijective function from <o1> to <o2>, assign a violation-mark for every undefined pair

5. Conclusion

To recapitulate the contribution of this work, we have adjusted the definition of Mparse in input-output relationship provided by McCarthy & Wolf (2005) to the domain of output-to-output correspondence. Because correspondence relation is domain-universal, a need for a bijective relationship judgment in both BR and OO correspondence was necessary. While implementing the same assessment mechanism, we have encountered several exemplary and stipulative issues. A common thread penetrating the problems was treating correspondence judgment binarily. This overlooked theoretical underpinnings of output-to-output relation in both practical and explanatory terms. By revising the constraint to be gradient, we could implement a more comprehensive understanding of output-to-output correspondence at the phonology-morphology interface. Last but not least, I would like to briefly note the potential benefits this revision could give to help our current theory surpass its competitors.

One alternative work to explain the phonological gap is the control model suggested by Orgun and Sprouse (1999). The centrality of this theory is to suggest a separate evaluative device CONTROL aside from EVAL. We have previously looked at how Mparse has a thresholding effect, eventually blocking out options violating the constraints above it. The control model characterizes this group under different settings, deeming it an insurmountable barrier. These constraints are removed from EVAL and taken to CONTROL. They are employed only at the second round of a judgment for pruning out the absolute impossibilities. The problem this theory faces with our current revision is that the strict division of CONTROL and EVAL no longer holds. In output-to-output correspondence, in particular, null parse can incur a higher violation cost than previously conceived. It can therefore not be the most optimal candidate among violators of the above-ranked constraints. In other words, as MParse-OO can no longer play the

gatekeeping role as it used to, building a control model on top of the *inviolable constraints >> MParse >> violable constraints* assumption will be unfitting. As a result, compared to binary viewing of the control, our revised O-O-MParse model will provide a more nuanced understanding of OO -correspondence.

Moreover, Rice (2005) suggests an optimal paradigm approach to address the inflectional gap in morphology. Instead of attributing the effect of phonological absence to the process within phonology, this approach directly places the problem at the phonology-morphology interface. By taking the entire morphological paradigm with gaps in different categories as input, MAX{category} constraint is responsible for ruling out specific categories that appear as gaps. The problem with this approach is its learnability with output-to-output correspondence such as stem-affix derivations. Unlike the inflectional paradigm, which the paper focuses on, the amount of affixes available for the derivation of the given word is infinite. Resultingly, a paradigm with an infinite number of candidates would enter the OT tableaux, making it no less convenient than postulating an additional theoretical device null parse and *Mparse, a superiority claimed by the paper. This complexity stands in contrast to the less complex indexing O-O-Mparse requires. In Mparse-IO, morphological context including case, plurality, and numbers resulted in various typologies. For output-to-output correspondence that engages more predetermined factors, the indexing of morphological context is simpler, especially compared to the infinite list of candidates in the optimal paradigm. For instance, the Mparse-BR would not need additional specification whereas the paradigm approach should enumerate all possible reduplicants that the given base can form.

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